EXPERIMENT 2: WATER JUG PROBLEM

Aim:

To develop a Python program that solves the Water Jug Problem using the State Space Search approach to obtain exactly 2 liters of water in the 3-liter jug.

Objective:

- To understand the Water Jug Problem in Artificial Intelligence.
- To perform operations such as filling, emptying, and pouring water between jugs.
- To obtain exactly 2 liters in the 3-liter jug.
- To demonstrate the sequence of states leading to the goal.

Algorithm:

Step 1: Start.

Step 2: Initialize both jugs as empty (4L = 0, 3L = 0).

Step 3: Fill the 3-liter jug completely.

Step 4: Pour water from the 3-liter jug into the 4-liter jug.

Step 5: Fill the 3-liter jug again.

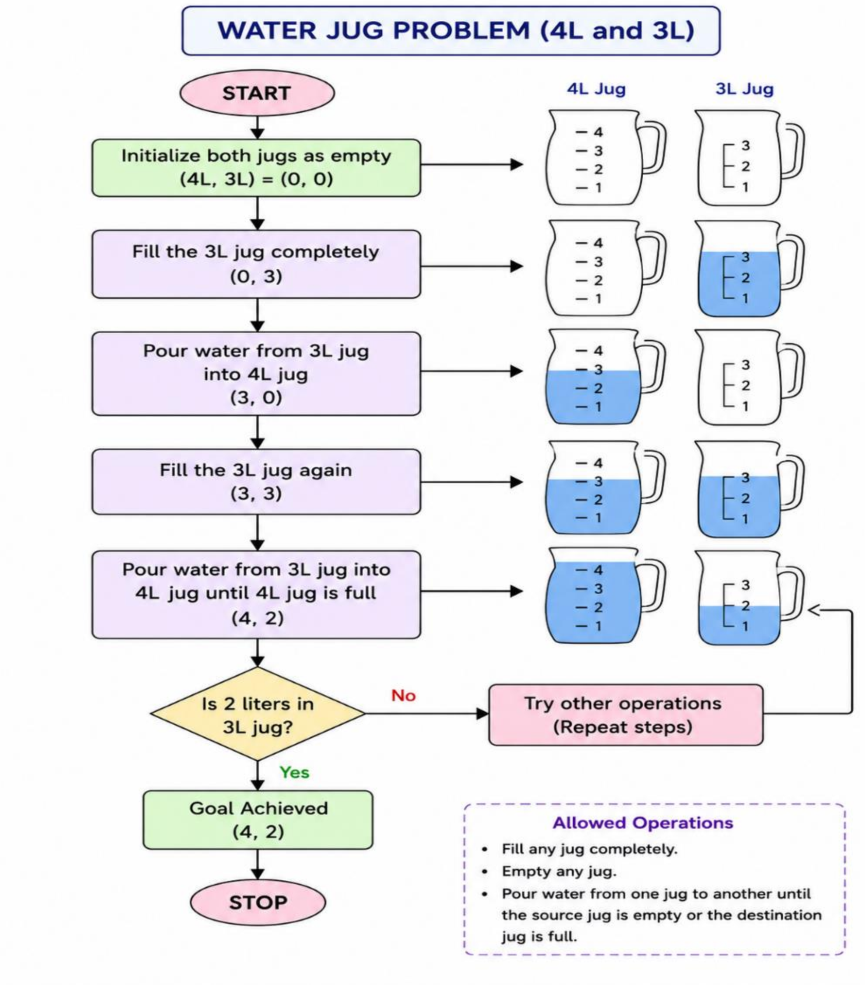
Step 6: Pour water from the 3-liter jug into the 4-liter jug until the 4-liter jug becomes full.

Step 7: The remaining water in the 3-liter jug is **2 liters**.

Step 8: Display the final state.

Step 9: Stop.

Flow chart:



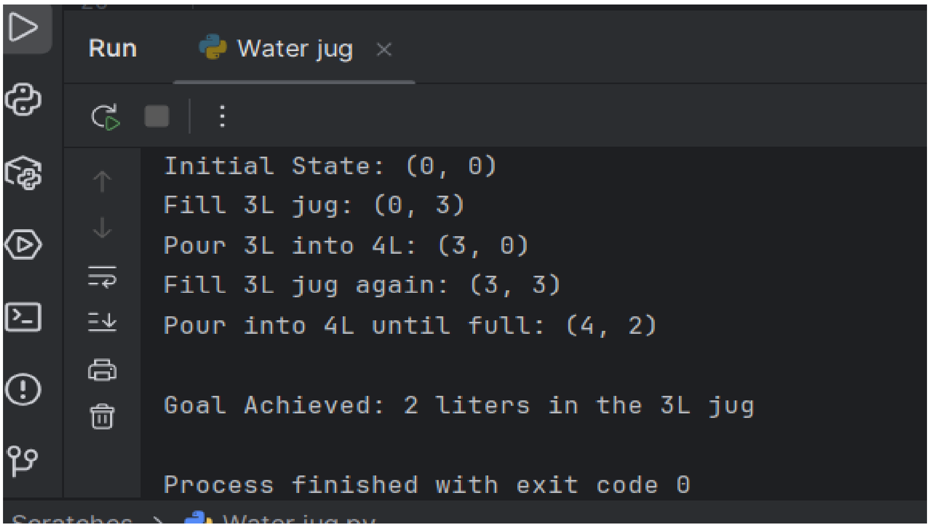
Source code:

```
# Water Jug Problem (4L and 3L)
jug4 = 0
jug3 = 0

print("Initial State:", (jug4, jug3))
# Fill 3L jug
jug3 = 3
print("Fill 3L jug:", (jug4, jug3))
# Pour 3L into 4L
jug4 = 3
```

```
jug3 = 0
print("Pour 3L into 4L:", (jug4, jug3))
# Fill 3L jug again
jug3 = 3
print("Fill 3L jug again:", (jug4, jug3))
# Pour from 3L to 4L until 4L is full
jug4 = 4
jug3 = 2
print("Pour into 4L until full:", (jug4, jug3))
print("\nGoal Achieved: 2 liters in the 3L jug")
```

Output:



Result :

The Python program successfully solves the Water Jug Problem and achieves the goal state with 2 liters of water in the 3-liter jug.